SHIVAM SHARMA

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SUMMARY

Target role: Robotics and AI Engineer. Analytical professional in autonomous systems, **Artificial Intelligence**, and **machine learning**, with expertise in **Python**, **LLMs**, and data-driven solutions. Experienced in agentic automation, robotics-focused ML pipelines, and intelligent system design.

EDUCATION

Master of Science in Robotics and Autonomous Systems (AI) — August 2025–May 2027
Arizona State University, Tempe, USA — **GPA: 3.9/4**
Relevant Coursework: **Robotics** Kinematics and Dynamics, Autonomous Navigation, **Deep Learning** for **Robotics**, **Computer Vision**, **Control Systems**, **Sensor Fusion**

Bachelor of Technology in Artificial Intelligence — August 2021–May 2025
Amity University, Noida, India — **GPA: 9.8/10**
Relevant Coursework: **Artificial Intelligence** Principles, **Machine Learning** Algorithms, Data Structures and Algorithms, Predictive Analytics, Neural Networks, Advanced Programming

EXPERIENCE

AI Agent Engineer, Remote Mechanical Tree Operator — February 2026–Present
Arizona State University (Global Futures Laboratory) — *Tempe, USA*
Tech Stack: Python, Multi-Agent AI Systems, Open-Source LLMs, GPT, Claude, Gemini, Agent Orchestration

- Architected and deployed a **multi-agent AI system** powered by a customized **open-source LLM** to autonomously control and optimize remote mechanical tree operations.
- Developed **AI-agent-driven data pipelines** to automate data collection, processing, and analysis from remote mechanical systems.
- Leveraged **GPT, Claude, and Gemini** for anomaly detection, insight generation, and **automated report generation** for research teams.

Machine Learning & AI Specialist — April 2022–June 2024
RSRTechno — *Noida, India*
Tech Stack: Python, Machine Learning, XGBoost, SMOTE, Optuna, SHAP, Streamlit, SQL

- Developed a scalable **churn prediction system** for 500K+ customers, applying **SMOTE** to improve high-risk identification precision by 25% annually.
- Engineered an end-to-end **ML pipeline** and optimized hyperparameters using **Optuna**, reducing deployment time by 20% for production teams.
- Implemented **Explainable AI (SHAP)** and **Streamlit dashboards** to visualize risk drivers, reducing manual review time by 18% and improving stakeholder decision-making.

TECHNICAL SKILLS AND CERTIFICATIONS

Programming Languages: Python
Machine Learning & Deep Learning: PyTorch, **TensorFlow**, scikit-learn
MLOps & Optimization: MLOps, **Optuna**
Computer Vision & Robotics: ROS (Robot Operating System), **Robotics**, **Control Systems**, **Sensor Fusion**
Natural Language Processing: Large Language Models (**LLMs**)
Data Automation Skills: Artificial Intelligence (**AI**), **Autonomous Systems**, Data Analysis, Data Processing, Data Visualization, **Deep Learning**, **Git**, **Machine Learning** (**ML**)
Soft Skills & Behavioural Keywords: Stakeholder Collaboration

ACADEMIC PROJECTS

Autonomous Mobile Robot Navigation System | ROS, **Python**, OpenCV, **Sensor Fusion**

- Developed an autonomous navigation system for mobile robots using **ROS**, achieving 90% path planning efficiency in template-driven environments.
- Implemented **sensor fusion** algorithms for improved obstacle detection, reducing collision incidents by 40% during simulated operations.
- Integrated **SLAM** techniques to enable real-time mapping and localization, enhancing robot positional **accuracy** to within 5 cm.

AI-Powered Anomaly Detection for Robotic Operations | Python, **TensorFlow**, scikit-learn, Data Visualization

- Engineered a **deep learning** model for proactive anomaly detection in **robotics** data, boosting detection **accuracy** to 96% with minimal false positives.
- Analyzed 1000+ hours of operational telemetry, identifying critical failure patterns to enhance system reliability by 15% through predictive alerts.
- Designed a Streamlit dashboard for visualizing anomalies and model predictions, enabling rapid root-cause analysis and operational decision-making.

Reinforcement Learning for Robotic Arm Manipulation | Python, **PyTorch**, OpenAI Gym, Gazebo

- Designed and trained a **reinforcement learning** agent using **PyTorch** to optimize robotic arm movements for pick-and-place tasks.
- Achieved a 92% success rate in simulated manipulation scenarios and improved **control precision** for industrial applications.
- Developed custom reward functions and explored policy gradient methods to enhance learning stability and task performance.